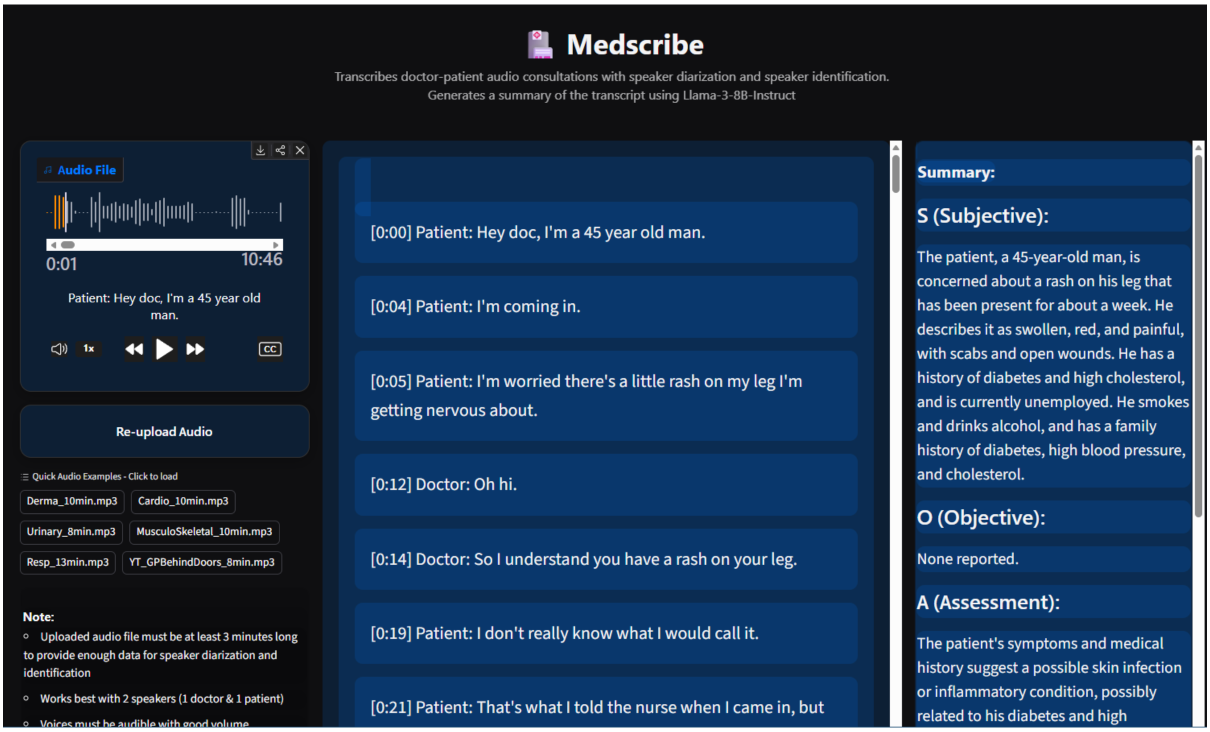
MedScribe



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Demo: https://huggingface.co/spaces/KroyEsp/MedScribe

Abstract

Medical report writing is a pain-point for medical doctors. MedScribe is an application that processes doctor-patient audio data to produce both clean medical transcripts and medical summary reports. Speaker diariazation was achieved using Resemblyzer embeddings and agglomerative clustering, while speaker identification was performed with a trained Bi-LSTM model that achieved 99% accuracy and recall scores. SOAP reports were generated by summarizing the transcription output using the Llama-3-8B Instruct LLM model. Therefore, by utilizing NLP tools and techniques, the application was able to address this pain-point experienced by doctors in both clinical and telemedical scenarios.

Contents

1. Introduction

Doctor-patient consultation is one of the fundamental parts of medicine. During this stage doctors, especially General Practitioners, need to assess the condition of their patient's health. Additionally, doctors are required to perform Electronic Health Record (EHR) tasks including writing SOAP notes (Subjective, Objective, Assessment, Plan) to summarize the consultation. However, writing medical reports for each patient is a tedious and time-consuming process. Doctors spend about six hours per day on EHR documentation tasks, while often needing to finish them outside of clinical hours (Arndt et al., 2017). Therefore, this is a significant pain-point for medical professionals.

The advancement of Artificial Intelligence tools and techniques, especially within the Natural Language Processing (NLP) domain, has the potential to address this problem. Speech Recognition Systems allows automatic transcription of the conversation between a doctor and patient. The generated transcript can be used as reference to make more informed clinical decisions. Additionally, abstractive summarization generated by Large Language Models (LLM), can quickly create medical reports based on the transcripts (Neupane et al., 2025). Thus, this can potentially reduce the time and effort required for the report writing process, allowing doctors to spend more time consulting patients instead.

The purpose of MedScribe is to implement both these functionalities into a single application. Thus, this application can provide doctors a practical tool that they can use in clinical settings as well as in telemedical consultations such as online or phone-based appointments.

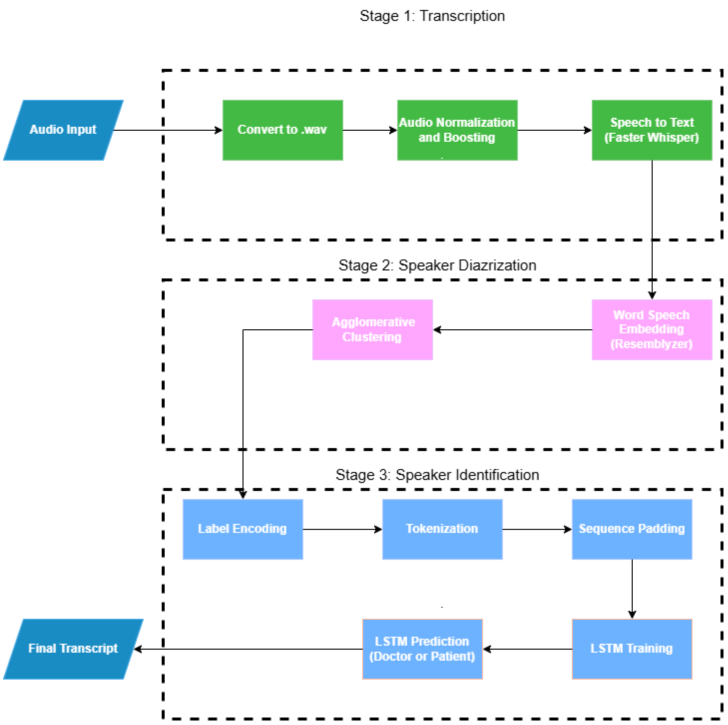
2. Methodologies

2.1 Speech Processing Pipeline

The application consists of three different stages for speech processing. The first stage transcribes the audio input. The second stage is speaker diarization which separates the audio data into parts corresponding to each individual speaker. This helps determine who spoke when in a conversation with multiple speakers. Speaker diarization is achieved by performing voice embeddings and clustering techniques. The last stage is speaker identification which classifies each speaker as either the doctor or patient using a Long-Short Term Memory (LSTM) model. The final output of the speech pipeline is a speech-segmented and labelled transcript.

Fig. 1

Speech processing pipeline of the application



2.2 Transcription

The transcription process involves pre-processing the audio input to improve the accuracy of both transcription and diarization. The audio format is converted into Waveform Audio File (WAV) format to ensure lossless audio quality. Normalization and audio boosting were applied to help better capture the voice data.

The Faster Whisper (medium-en) model was used to transcribe the text. This model is both lightweight and faster than the base Whisper model while still generating accurate transcriptions. It can also extract word-level timestamps which is essential for the embedding process of speaker diarization. Furthermore, the model includes a Voice

Activation Detection (VAD) filter to exclude non-speech parts, which helped minimize audio hallucinations.

2.3 Speaker Diarization

2.3.1 Voice Embeddings

Resemblyzer was used to extract high-level voice features by embedding each spoken word from the audio into a vector representation (Jemine, 2023). It uses a pre-trained deep learning model to encode the voice into a vector embedding of 256 values. This embedding captures each speaker's voice characteristics and is used as the input for the clustering algorithm.

2.3.2 Clustering

Agglomerative clustering was used to segment the extracted embeddings into speaker-specific groups or clusters. It is a type of Hierarchical clustering which iteratively merge similar clusters until a single cluster remains. It is commonly used in speaker differentiation because of its advantage of forming clusters without specifying a predefined K-value (Banu & Deshmukh, 2026). Instead, a distance threshold can be used to determine when clusters should merge. Hence, this allows diarization on audio files with unknown number of speakers.

2.4 Speaker Identification

2.4.1 Dataset

The dataset by Ahmed (2024), obtained from Kaggle, was used to train the LSTM model. This dataset contains both audio and text transcript of a simulated doctor-patient consultation. It consists of 544 labelled transcripts from a small variety of medical consultation topics including Cardiology, Respiratory and Musculoskeletal.

Fig. 2

Example labelled transcript with the doctor labelled as 'D' and patient labelled as 'P' *(*Ahmed, 2024)

D: It sounds like that you're experiencing some chest pain.

P: Yeah, so this chest pain has just been coming on for the last couple of weeks now.

D: OK, uhm. So 2 weeks, and can you tell me kind of where you're feeling that pain?

P: Um yeah, I'm I'm just feeling this chest discomfort, sort of over on the left side.

D: The left side, OK. Does does it stay just in one spot or does it kind of spread anywhere else?

P: No, I don't really feel it anywhere else. Yeah, I just feel it in that one spot on the left side.

D: OK, does it feel dull and achy, or does it feel more sharp?

P: It feels pretty sharp.

D: OK, and is it constant like on all the time, or do you feel like it kind of comes and goes?

P: It comes and goes. I would say I it might last for um 20-30 minutes at a time and then it goes away.

D: OK, and over the last two weeks since you said it started, has it gotten any worse?

P: Uh, no. It's been about the same.

D: About the same, OK, and has it gotten more frequent, these like 20 to 30 minute episodes?

P: No, it's happened about two or three times during the 2 weeks.

2.4.2 LSTM model

LSTM models are a Recurrent Neural Network (RNN) that are commonly used for speaker identification across various applications, such as in air traffic communications (Guo et al., 2023). This is due to their ability to capture long-term dependencies especially in lengthy conversations. Furthermore, a Bi-Directional LSTM model can better capture semantic patterns in conversational data by considering both the preceding and subsequent texts (Guo et al., 2023). Hence, a Bi-LSTM is a suitable model for classifying each speaker as either the doctor or the patient, because it can leverage the bi-directional conversational context.

The Bi-LSTM model consists of a single bi-directional layer with 32 hidden units, followed by a Dense layer with 32 units using the ReLu activation function. Dropout layers, with a rate of 0.5, and Batch Normalization were added between each layer to prevent the model from overfitting. The output layer is a Dense layer with the Sigmoid activation function. Adam optimizer with a learning rate of 0.001 was used, and the model was trained for a maximum epoch of 15 with a batch size of 16. Early stopping was used to minimize overfitting.

2.5 Abstractive Summarization

Abstractive summarization was chosen because generating a SOAP medical report requires paraphrasing the transcription output rather than simply extracting relevant parts. This was achieved by using Meta's Llama-3-8B Instruct LLM model. This model is optimized for dialogue-based conversations and can perform summarization and rewriting tasks effectively when given a specific instruction or prompt (Meta, 2024). Therefore, it is a suitable model for generating medical report summaries from the transcription output.

Fig. 3

Instruction prompt provided to the Llama LLM model

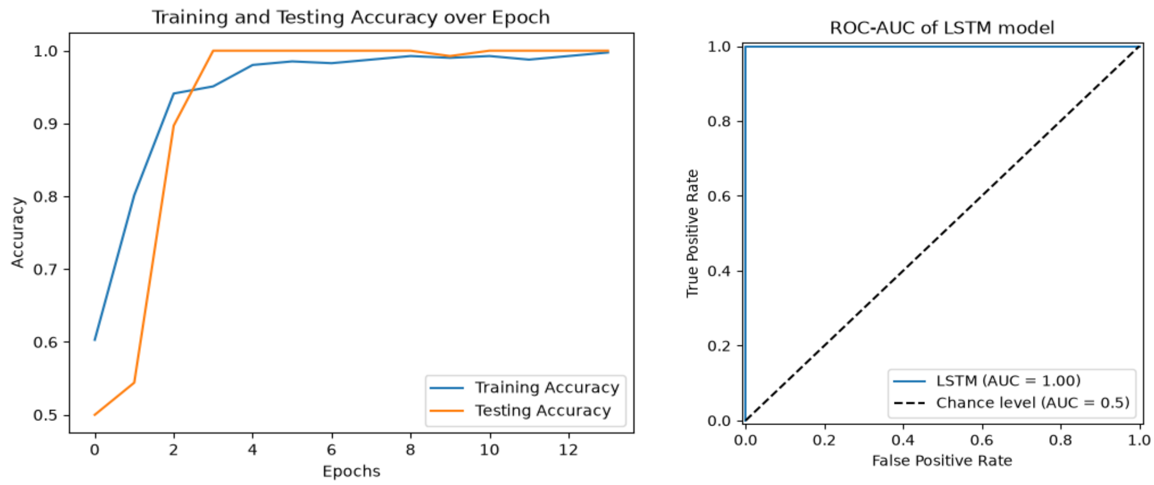
```
Generate a SOAP-style clinical summary based on the given Conversation between Doctor and Patient.

S (Subjective): Summarize the patient's subjective concerns, symptoms, context, and relevant history.
O (Objective): Extract objective findings mentioned in the transcript (vitals, observations, exam findings, measurable data).
A (Assessment): Provide a concise assessment of the patient's condition based only on the transcript.
P (Plan): Suggest an appropriate plan or next steps based strictly on what was discussed.

If a section has no information, write "None reported." DO not include the instructions in the output.
Format the final answer exactly as:
# S (Subjective):
[subjective content]

# O (Objective):
[objective content]

# A (Assessment):
[assessment content]

# P (Plan):
[plan content]

Conversation:
{full_transcript}
```

3. Result

The training accuracy plateaued at 100% accuracy after approximately 8 epochs while the testing accuracy plateaued after around 3 epochs (Figure 4). This suggests that the Bi-LSTM model quickly generalised with the given dataset. Additionally, the gap between the training and testing is not wide which suggests no major overfitting. The model achieved an accuracy, recall, precision and F1 score of 99%. Furthermore, the AUC-ROC value was 1.0. Overall, the trained Bi-LSTM model exhibited high performance with the given dataset.

Fig. 4

Performance scores of the Bi-LSTM model



Performance Scores of LSTM

	Accuracy	Recall	Precision	F1-Score
LSTM	99.264706	99.264706	99.275362	99.264666

4. Discussion

The results of the LSTM model have shown that speaker identification can be achieved solely from textual content. This is because of the distinct linguistic and textual patterns that can reliably differentiate doctors from patients (Zolensky et al., 2025). For example, doctors are more likely to use medically related vocabulary and frequently use the pronoun “we” followed by auxiliary verbs, whereas patients are more likely to use the pronoun “I” (Skelton et al., 2002). Additionally, a study by Zolnoori et al. (2023) has shown that clinicians are more likely to ask formal questions while patients tend to use more informal language. Hence, the LSTM model was able to learn and classify these textual patterns effectively for speaker identification.

4.1 Limitations

One of the major limitations is the reduced speaker diarization accuracy with higher number of speakers. The application performs best in one-on-one consultation between a doctor and a patient, which is the most common clinical scenario. However, in cases where multiple doctors or family members are present, the voice characteristics and linguistic patterns are highly likely to be similar, making it difficult for the agglomerative clustering algorithm to form distinct clusters and thus resulting in poorer transcription output. Additionally, overlapping speeches and excessive background noise can further reduce the performance.

Hence, it is recommended that the audio file is at least five minutes in length to provide sufficient data to accurately differentiate the speakers

Another limitation is that the Bi-LSTM model training was limited to a small, single dataset. This may explain the extremely high model performance as the contextual speech patterns can be relatively consistent and homogeneous throughout this single dataset. As a result, the model is highly likely to poorly generalize in real-world scenarios, especially since doctors can communicate with their patients in very different ways.

4.2 Challenges

One of the main challenges encountered was the heavy computational requirement required to process both transcription and summarization. Initially, the summarization process was done using a local version of the Llama LLM model. This was a significant performance bottleneck as running a local LLM model locally is extremely computational taxing.

To tackle this issue, the API version was used instead. As a result, the application's processing speed was improved, being able to process a 10-minute audio input in approximately 1-2 minutes. However, a disadvantage of using the API version is the additional cost, although the Llama LLM model has a very low cost per token.

5. Conclusion

Speaker diarization, achieved through a combination of Resemblyzer and agglomerative clustering, together with speaker identification, produced clean and easy-to-read medical transcripts. The trained Bi-LSTM model was able to identify each role with high performance, without relying on vocal features. The Llama LLM model successfully generated a SOAP medical summary based on the transcription output. Therefore, MedScribe was able to process the doctor-patient speech data and produce text outputs that are potentially beneficial for doctors.

To further improve the application's speaker identification accuracy, a LLM approach can be considered to better understand the semantic patterns between each role. A fine-tuned BERT model has been shown to effectively capture these role-specific semantic cues directly from long transcripts, achieving high accuracies even without applying speaker diarization beforehand (Zolensky et al., 2025).

Improving the medical SOAP summary generated by the LLM model involves reducing textual hallucinations and mitigating their effects through several considerations. First, is to use a fine-tuned LLM model that is specifically trained within medical domain. MedGemma is an example of a suitable domain-specific model that is designed to generate clinical text (Sellergren et al., 2026). Secondly, a Retrieval-Augmented Generation (RAG) system can be implemented to reduce hallucinations by referring the text output to a vector database containing medical transcripts with a corresponding SOAP report (Neha et al., 2025). Lastly, an Epistemic Neural Network can be developed to estimate the uncertainty of the generated output (Osband et al., 2023). By highlighting parts of the generated text with high uncertainty, doctors are more likely to pay more attention on these areas and correct any potential misinformation.

Overall, incorporating these improvements enables the application to be used as a reliable support tool that improves not only the efficiency but also the accuracy of the doctor-patient consultation process.

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